

Project Proposal: Pathological Myopia Diagnosis Using AI

Project Title:

AI-Powered Pathological Myopia Diagnosis System

2. Project Overview:

Pathological myopia is a severe form of myopia (nearsightedness) that can lead to vision loss due to structural changes in the eye. The goal of this project is to develop an AI-powered system that can accurately diagnose pathological myopia using retinal images. The system will leverage machine learning and image processing techniques to assist healthcare professionals in early detection, helping to reduce the risk of irreversible vision damage.

Problem Statement:

Traditional methods for diagnosing pathological myopia rely heavily on subjective evaluation by specialists, making it time-consuming and prone to human error. There is a need for an automated diagnostic tool that can analyze retinal images quickly and accurately to provide a reliable second opinion and support timely interventions.

Objectives:

1. Develop an AI-based diagnostic tool for identifying pathological myopia using retinal images.
 2. Collect and preprocess a diverse dataset of retinal images from online sources (e.g., Kaggle, Kargo) as well as from previous patients of eye specialists.
 3. Train a machine learning model using image classification techniques.
 4. Evaluate the model's performance using metrics such as accuracy, precision, recall, and F1-score.
 5. Create a user-friendly interface for healthcare professionals to upload images and receive diagnostic results.
 6. Document and validate the results to ensure clinical applicability.
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Methodology:

1. Data Collection:

- Gather retinal images from publicly available datasets (e.g., Kaggle, Kargo) and categorize them into normal and pathological myopia cases.

2. Data Preprocessing:

- Perform image augmentation, normalization, and resizing.
- Use techniques like image segmentation to highlight relevant features.

3. Model Development:

- Train Convolutional Neural Networks (CNN) or similar architectures to classify the images.
- Experiment with transfer learning models such as ResNet or VGGNet for improved accuracy.

4. Evaluation:

- Use a test dataset to evaluate model performance.
- Fine-tune hyperparameters based on validation results.

5. Deployment:

- Implement a web-based and mobile-based interface to enable real-time image analysis.
- Integrate feedback and testing from healthcare professionals for further refinement.

6. User Testing and Validation:

- Conduct usability testing with target users (e.g., ophthalmologists).
- Validate results through comparison with clinical evaluations.

Deliverables:

1. A trained and validated machine learning model capable of diagnosing pathological myopia.
2. A web-based tool for uploading and analyzing retinal images.
3. A mobile-based application for ease of access.
4. Documentation covering the development process, evaluation, and results.
5. A final project report and a presentation of findings.

Project Timeline:

Phase	Activities	Duration
Phase 1: Requirements	Requirement gathering and research	1 week
Phase 2: Data Collection	Collection and preprocessing of retinal images	2 weeks
Phase 3: Model Development	Training and fine-tuning machine learning models	3 weeks
Phase 4: Testing and Evaluation	Model evaluation and validation with test data	2 weeks
Phase 5: Deployment	Building and deploying the application	2 weeks
Phase 6: User Testing	Collecting feedback and refining the tool	1 week
Phase 7: Finalization	Documentation and final project report	1 week

Required Resources:

- **Hardware:** High-performance computing resources for training models.
- **Software:** Python, TensorFlow/PyTorch, OpenCV, React (for frontend), Flask/Django (for backend).
- **Datasets:** Retinal images from sources like Kaggle or Kargo and from eye specialists.

- **Human Resources:** AI/ML engineers, software developers, and domain experts (e.g., ophthalmologists).
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Expected Outcomes:

1. A robust AI model for detecting pathological myopia.
 2. A functional diagnostic tool that can support medical professionals in early identification of the condition.
 3. Enhanced understanding of the use of AI in healthcare.
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LITERATURE REVIEW

The integration of artificial intelligence (AI) in the diagnosis of pathological myopia has garnered significant attention in recent years, reflecting a growing recognition of the potential for machine learning and deep learning techniques to enhance diagnostic accuracy and efficiency.

Pathological myopia, characterized by excessive axial elongation of the eye leading to structural changes, poses a considerable challenge in clinical settings due to its complex nature and the subjective variability in traditional diagnostic methods (Zhang et al., 2013; Kim et al., 2021).

Several studies have explored the application of AI in diagnosing pathological myopia, with promising outcomes. For instance, Zhang et al. proposed a computer-aided diagnosis framework that utilizes multiple kernel learning (MKL) methods to fuse heterogeneous biomedical data, significantly improving diagnostic accuracy (Zhang et al., 2013). Similarly, Kim et al. demonstrated the effectiveness of machine learning algorithms in predicting pathological myopia by analyzing tomographic elevation of the posterior sclera, establishing a correlation between topographic features and disease classification (Kim et al., 2021; Ko et al., 2022). These findings underscore the potential of AI to leverage diverse data sources for enhanced diagnostic precision.

Zhang et al. (2023) conducted a systematic review and meta-analysis to evaluate the performance of AI-based models in detecting pathologic myopia (PM) and its complications using fundus and OCT images. A total of 22 studies were reviewed, with 14 included in the quantitative analysis. The pooled sensitivity and specificity for detecting PM were 95% and 97%, respectively, and the area under the curve (AUC) was 0.99, indicating high diagnostic accuracy. The study highlighted AI's potential for effective PM detection and the need for standardized methodologies across research. Limitations noted include variability in dataset characteristics and image sources, which affect algorithm generalizability.

Deep learning approaches have also been pivotal in advancing the automated detection of pathological myopia. Hemelings et al. reported the successful application of deep learning models for lesion segmentation and classification using fundus images, achieving high performance metrics such as area under the receiver operating characteristic curve (AUC) (Hemelings et al., 2021). Furthermore, Devda and Eswari developed a convolutional neural network (CNN) specifically for pathological myopia image analysis, which yielded satisfactory results in both classification and segmentation tasks (Devda & Eswari, 2019; Wang et al., 2022). These studies illustrate the capability of deep learning to not only detect pathological changes but also to provide detailed insights into the specific lesions associated with the condition.

The establishment of comprehensive datasets has further facilitated research in this domain. The PALM dataset, comprising 1200 annotated fundus images, has been instrumental in training and evaluating AI models for pathological myopia recognition (Fang, 2024). This dataset enables researchers to develop and benchmark their algorithms against a standardized set of images, thereby enhancing the reproducibility and reliability of their findings. Additionally, the use of transfer learning techniques has been highlighted as a means to optimize model performance, particularly when training data is limited (Xie, 2019; Chen et al., 2021). Despite the advancements, challenges remain in the clinical implementation of AI for pathological myopia diagnosis. Issues such as the need for external validation datasets and the integration of AI systems into existing clinical workflows are critical for ensuring the trustworthiness and efficacy of these technologies (Tang et al., 2022). Nonetheless, the body of literature indicates a clear trajectory toward the incorporation of AI in ophthalmology, with the potential to revolutionize the diagnostic landscape for pathological myopia and improve patient outcomes.

The exploration of artificial intelligence (AI) in diagnosing and managing pathological myopia in Uganda is under-researched. However, discussions on AI in healthcare across Africa are emerging and may offer insights into its potential for myopia diagnosis. Nibigira's study highlights challenges like infrastructural deficits and inadequate governance, which hinder AI adoption in African countries, including Uganda. Haarman et al.'s meta-analysis emphasizes the need for better diagnostic tools globally, suggesting AI's potential. Babarinde's review

compares AI adoption in the USA and Africa, stressing the need for local capabilities. Zhang and Zou's work underscores AI's relevance in addressing myopia worldwide. Although there are no direct studies on Uganda, the literature suggests AI's potential in enhancing myopia diagnosis, contingent on overcoming infrastructural and governance barriers.

In conclusion, these literatures reveals a concerted effort to harness AI for the diagnosis of pathological myopia, with various studies demonstrating the effectiveness of machine learning and deep learning methodologies. The outcomes of these investigations suggest that AI can significantly enhance diagnostic accuracy, streamline workflows, and ultimately contribute to better management of this complex condition.

NOTE: Despite the extensive research and advancements in AI for pathological myopia diagnosis, there is a noticeable gap in studies specifically conducted by undergraduate students, particularly from Makerere University. Our review did not find any documented undergraduate-level research addressing this problem. This highlights the unique contribution of our project, as it seeks to demonstrate the capability of undergraduate students to tackle complex issues in medical diagnostics using AI, potentially paving the way for similar future studies at the undergraduate level.

Differentiation from Previous Approaches

The existing body of research has established a strong foundation for using AI to diagnose pathological myopia, leveraging diverse machine learning models and imaging techniques. Zhang et al. (2013) and Kim et al. (2021) applied machine learning algorithms to fuse heterogeneous data and analyze tomographic elevation, respectively, which improved diagnostic accuracy but focused primarily on enhancing the classification performance of already known structural features. Similarly, deep learning models developed by Hemelings et al. (2021) and Devda & Eswari (2019) emphasized lesion segmentation and image classification but faced challenges in real-world implementation due to limited generalizability across diverse patient populations.

Unlike these studies, our approach aims to combine AI with a context-sensitive framework that considers demographic and geographic variations specific to Uganda, where AI adoption in healthcare is nascent (Nibigira, 2024). We propose a hybrid model that incorporates both clinical data and socio-demographic variables, enabling a more holistic understanding of pathological myopia progression in under-researched regions. Additionally, while previous studies have focused on standard datasets like PALM (Fang, 2024), we will utilize locally sourced data to train and validate our models, addressing concerns about data representativeness and model robustness highlighted in global reviews (Babarinde, 2023).

By adopting a context-aware design and focusing on data diversity, our approach not only seeks to replicate high diagnostic accuracy but also to demonstrate the feasibility of implementing AI-driven solutions in resource-constrained settings. This differentiation addresses the infrastructural challenges noted by Nibigira (2024) and positions our research as a stepping stone toward more localized AI healthcare applications in Uganda.

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Conclusion:

This project aims to build an innovative diagnostic tool using AI to aid in the early detection of pathological myopia. It is anticipated that this solution will contribute to improving clinical outcomes and reducing vision loss due to late-stage diagnosis.